**DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS**

# Mathematics III Topics in Analysis (3TN)

**Tutorial Sheet 4** **15 August 2024**

1. Let $V = \mathcal{C}([0,1],\mathbb{R})$ be the space continuous functions from $[0,1]$ to $\mathbb{R}$ equipped with the norm
$$\|f\|_1 = \int_0^1 |f(t)|dt \quad \forall f \in V\,.$$
Let $L_0 : V \to \mathbb{R}$ be defined by $L_0(f) := f(0) \ \ \forall f \in V$.
   (b) Show that $L$ is a linear mapping.
   (b) Show that $L_0 \notin V'$.
   (c) What does this show for linear mappings involving infinite dimensional normed spaces?

2. Let $\ell^1 := \{x = (x_n)$ sequence of real numbers such that $\sum_{n=0}^{\infty} |x_n| < +\infty\}$ be equipped with the norm
$$\|x\|_1 := \sum_{n=0}^{\infty} |x_n| \qquad \forall x = (x_n) \in \ell^1\,.$$
Let $m \in \mathbb{N}$ and set $L_m(x) := \sum_{n=0}^{m} x_n$ for every $x = (x_n) \in \ell^1$. Show that $L_m \in (\ell^1)'$, the dual space of $\ell^1$ and that $\|L_m\|_{(\ell^1)'} = 1$.

3. Let $V = \mathcal{C}^1([a,b],\mathbb{R})$ be the vector space over $\mathbb{R}$ of functions continuous with their first dervative from $[a,b]$ to $\mathbb{R}$. We consider the mapping $\|\cdot\|_* : V \to \mathbb{R}$ defined by
$$\|f\|_* := |f(a)| + \int_a^x |f'(t)|dt \quad \forall f \in V\,.$$
   (a) Show that $\|\cdot\|_*$ is a norm on $V$.

   (b) Consider the mapping $T : V \to V$ by
$$T(f)(x) = \int_a^x f(t)dt \quad \forall x \in [a,b], \quad \forall f \in V\,.$$
      (i) Show that $T$ is well-defined i.e., $T(f) \in V$ for every $f \in V$.
      (ii) Show that $T \in \mathcal{L}(V,V)$ with $\|T\|_{\mathcal{L}(V,V)} \leq b - a$.
      (iii) Show that $\|T\|_{\mathcal{L}(V,V)} = b - a$.

4. Let

$$V := \Big\{ f \in \mathcal{C}([0,1],\mathbb{R}) \colon \exists C(f) \geq 0 \,|\, |f(x)| \leq C(f)x \quad \forall x \in [0,1] \Big\}$$

(a) Show that $V$ is linear subspace of $\mathcal{C}([0,1],\mathbb{R})$.

(b) Prove that the mapping $\|\cdot\|_V : V \to [0,\infty[$ defined by

$$\|f\|_V := \sup_{0<x\leq 1} \frac{|f(x)|}{x} \qquad \forall f \in V$$

is a norm on $V$.

(c) For any $f \in V$ set $L(f)(x) := \begin{cases} \dfrac{1}{x}\displaystyle\int_0^x f(y)dy & \text{if } x \in (0,1] \\ 0 & \text{if } x = 0\,. \end{cases}$.

(i) Prove that $L(f) \in V$ for any $f \in V$. (This means that the mapping $L$ is well-defined).

(ii) Prove that $L \in \mathcal{L}(V,V)$ with $\|L\|_{\mathcal{L}(V,V)} \leq \dfrac{1}{2}$.
You can write $\displaystyle\int_0^x f(y)dy = \int_0^x y\frac{f(y)}{y}dy$.

(iii) Using the function $f(x) = x$, deduce that $\|L\|_{\mathcal{L}(V,V)} = \dfrac{1}{2}$.

5. Let $V = \mathcal{C}^1([0,1],\mathbb{R})$ be the space of continuously differentiable functions from $[0,1]$ to $\mathbb{R}$.

(a) Say whether the space $(V,\|\cdot\|_\infty)$ is a Banach space. (Hint: consider the sequence of functions $(f_n)$ defined by $f_n(x) := \sqrt{x^2 + \dfrac{1}{n+1}}$ $\forall x \in \mathbb{R},\ \ \forall n \in \mathbb{N}$.)

(b) Consider the mapping $N : V \to \mathbb{R}$ defined by $N(f) := \|f\|_\infty + \|f'\|_\infty$ $\forall f \in V$.

(i) Show that $N$ is a norm on $V$.

(ii) Show that $(V,N)$ is a Banach space. (Hint: Let $(f_n)$ be a Cauchy sequence in $(V,N)$. Deduce that $(f_n)$ and $(f_n')$ are Cauchy sequences in $(\mathcal{C}[0,1],\mathbb{R}),\|\cdot\|_\infty)$ (which is known to be complete) and hence, converge to $f$ and $g$ respectively in the norm $\|\cdot\|_\infty$. Then, write $f_n(x) = f_n(0) + \int_0^x f_n'(t)dt$ and pass to the limit. Then, deduce that the function $f$ is differentiable with $f'(x) = g(x)$ and conclude that $N(f_n - f) \to 0$.)

6. (a) Let $(V,+,\cdot)$ be a vector space over the field $\mathbb{K} := \mathbb{R}$ or $\mathbb{C}$. Define what is meant by the statement that two norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on $V$ are equivalent.

(b) Let $\|\cdot\|_1$ be the norm on $\mathbb{R}^2$ defined by $\|(x,y)\|_1 := |x| + |y| \quad \forall (x,y) \in \mathbb{R}^2$. Let $N : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$N(x,y) := \sup_{t\in[0,1]} |x + ty| \quad \forall (x,y) \in \mathbb{R}^2\,.$$

Find two constants $C_1$, $C_2 > 0$ such that

$$C_1 \|(x,y)\|_1 \leq N(x,y) \leq C_2 \|(x,y\|_1 \quad \forall (x,y) \in \mathbb{R}^2.$$